



DATASET OVERVIEW

Dataset Used

- Apple Inc. (AAPL) Historical Stock Data
- Period: June 2021 – June 2026
- Features: Date, Open Price, Close Price, High Price, Low Price, and Trading Volume



PYTHON TECHNIQUES APPLIED

- Data Cleaning and Preprocessing
- Handling Missing Values
- Date-Time Manipulation
- Feature Engineering
- Grouping and Aggregation
- Data Transformation using Pandas



CLOSING PRICE TREND

How has Apple's Stock Price Changed Over the Last 5 Years?



Insight:

- Apple's stock price generally increased from 2021 to 2026 despite temporary declines, showing strong long-term growth.

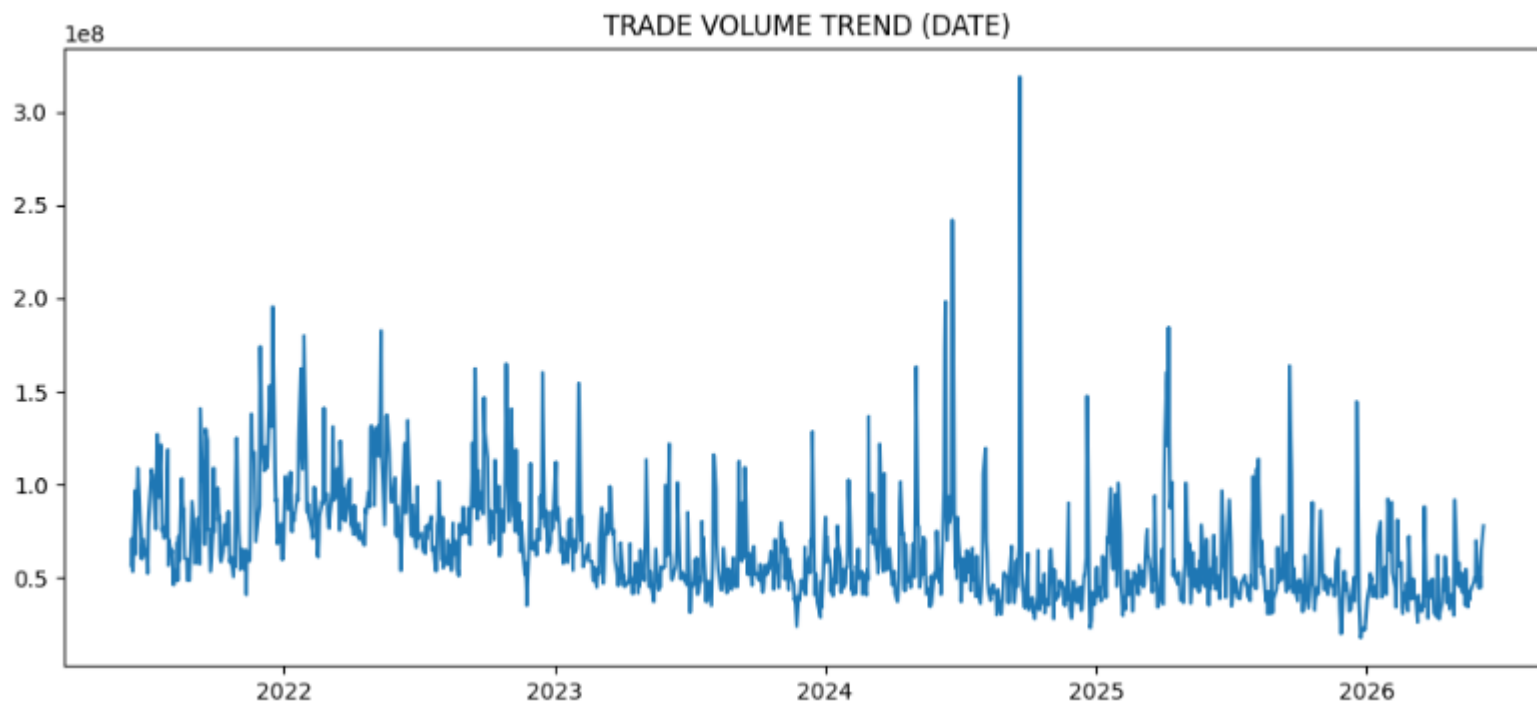
Recommendation:

- Long-term investors may benefit from a buy-and-hold strategy as the stock maintained an overall upward trend.



TRADING VOLUME TREND

How Active Were Investors In Trading Apple Shares?



Insight:

- Trading volume fluctuated throughout the period, but the stock price continued to rise, indicating sustained investor confidence.
- Daily volume mostly ranged between 50M–100M shares.
- Several spikes exceeded 200M shares.

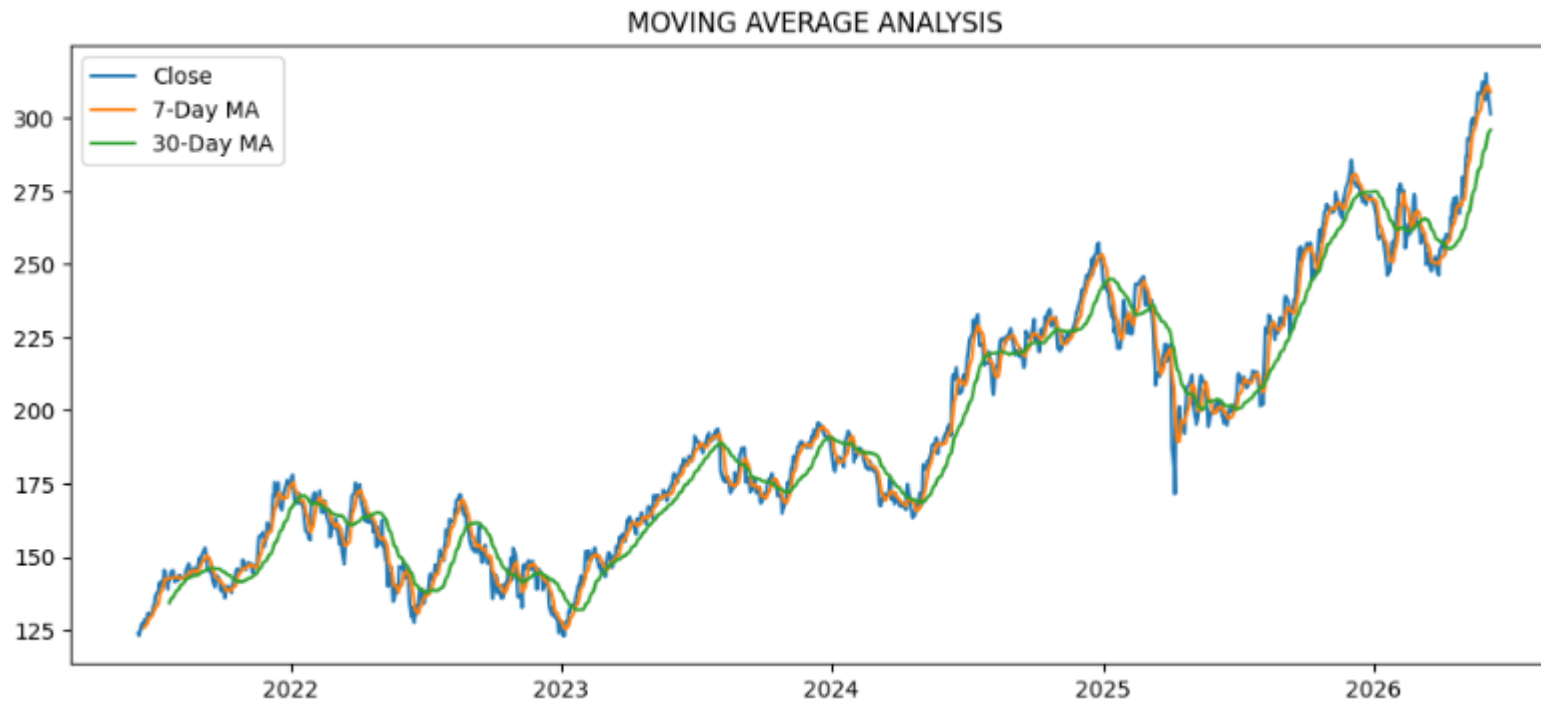
Recommendation:

- Investigate dates with unusual volume spikes. They may be linked to major company or market events.



MOVING AVERAGE ANALYSIS

What is the Overall Direction of Apple's Stock?



Insight:

- The 7-Day and 30-Day Moving Averages trended upward, confirming a strong long-term bullish trend despite temporary periods of weakness like in 2022 and 2025

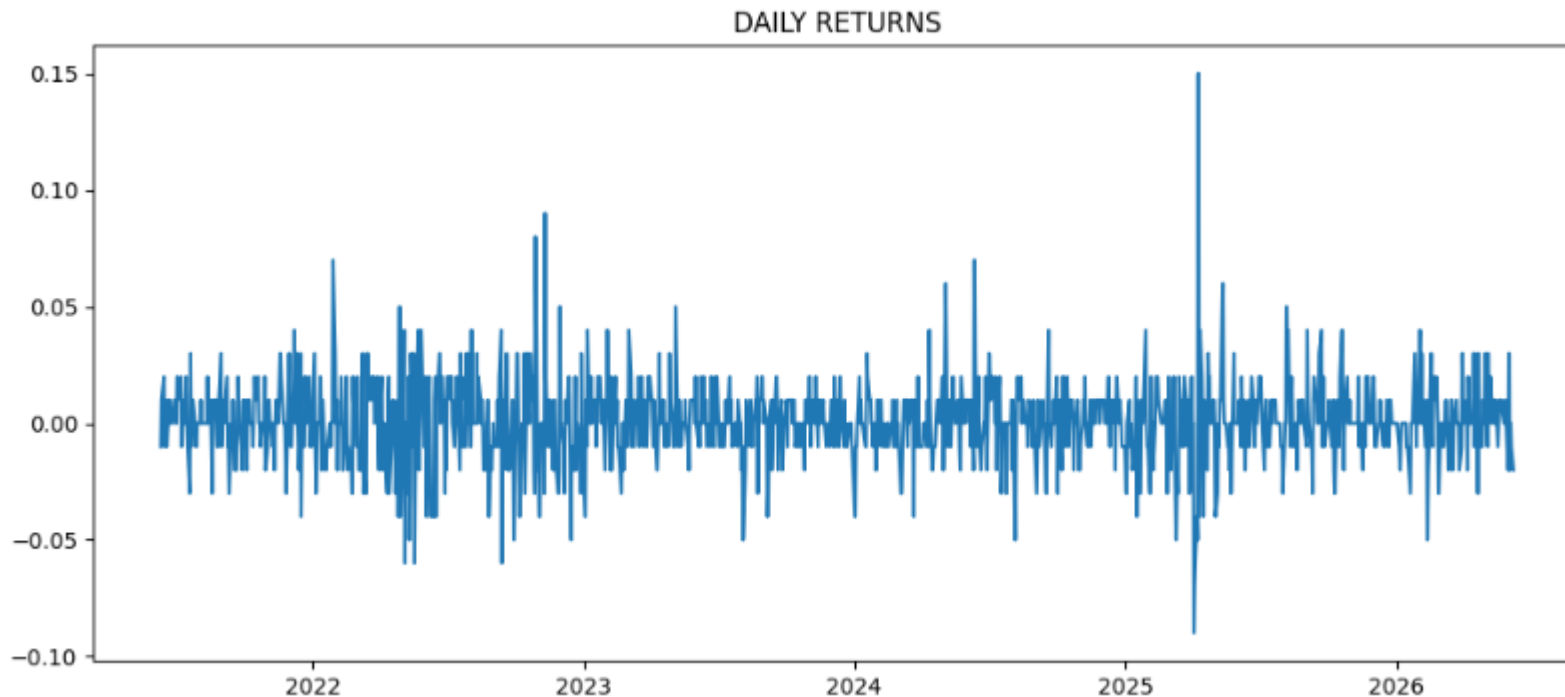
Recommendation:

- Focus on the long-term trend rather than reacting to short-term price fluctuations.



DAILY RETURN ANALYSIS

How much did Apple's stock gain or lose on a daily basis?



Insight:

- Daily returns fluctuated around zero, with most gains and losses remaining relatively small, indicating stable performance.

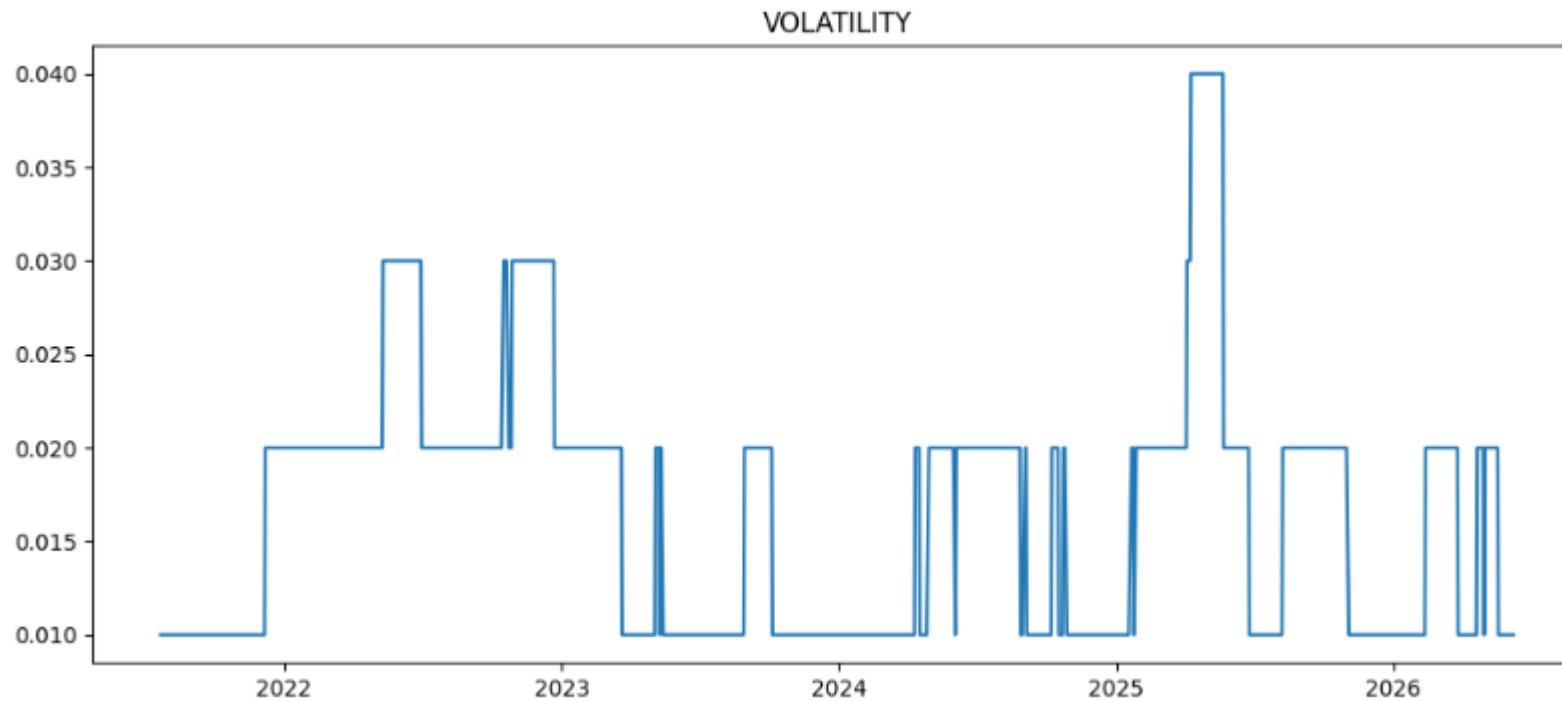
Recommendation:

- Avoid decisions based on a single day's return. Consider volume and market news alongside returns.



VOLATILITY ANALYSIS

How stable or risky was Apple's stock over time?



Insight:

- Volatility remained moderate for most periods.
- A few spikes occurred, especially around 2025.
- The stock maintained its long-term upward trend.

Recommendation:

- Pay closer attention during periods of high volatility, as they may signal increased risk and uncertainty.



KEY FINDINGS AND RECOMMENDATION

Key Findings

- Apple maintained a strong long-term upward trend.
- Investor confidence remained strong despite fluctuations in trading activity.
- Moving averages confirmed sustained growth.
- Daily returns showed generally stable performance.
- Volatility spikes were temporary and did not change the overall trend.

Recommendation

- Use a combination of price trends, volume, moving averages, daily returns, and volatility when evaluating stock performance rather than relying on a single indicator.